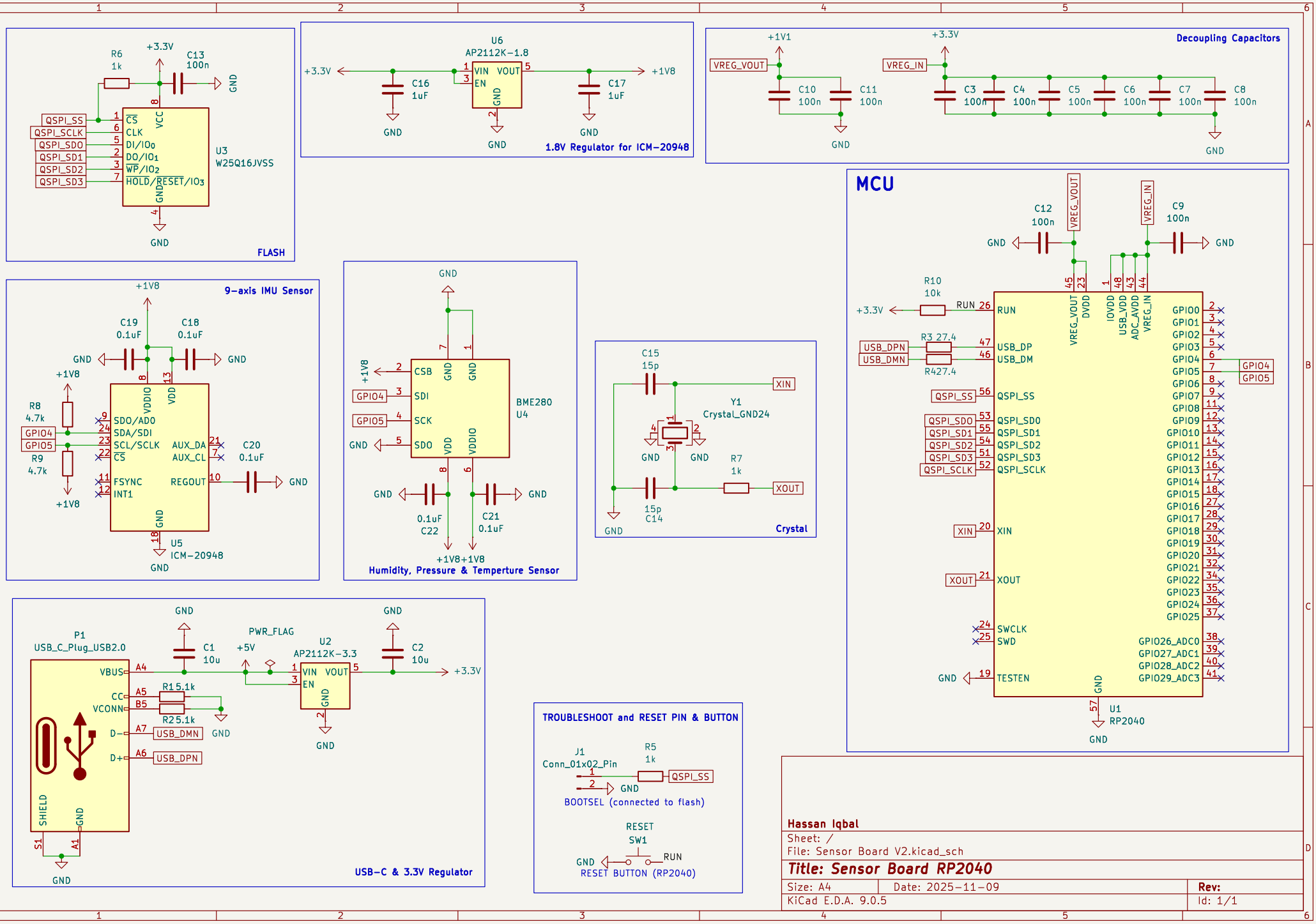

FLASH
R6
1k
+3.3V
C13
100n
GND
QSPI_SS
QSPI_SCLK
QSPI_SDO
QSPI_SD1
QSPI_SD2
QSPI_SD3
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
U3
W25Q16JVSS
1.8V Regulator for ICM-20948
U6
AP2112K-1.8
VIN
VOUT
EN
C16
1uF
C17
1uF
+1V8
Decoupling Capacitors
+1V1
VREG_VOUT
VREG_IN
C10
100n
C11
100n
C3
100n
C4
100n
C5
100n
C6
100n
C7
100n
C8
100n
9-axis IMU Sensor
C19
0.1uF
C18
0.1uF
R8
4.7k
R9
4.7k
GPIO4
GPIO5
SDO/ADO
SDA/SDI
SCL/SCLK
CS
FSYNC
INT1
VDDIO
VDD
AUX_DA
AUX_CL
REGOUT
C20
0.1uF
U5
ICM-20948
Humidity, Pressure & Temperture Sensor
CSB
SDI
SCK
SDO
BME280
U4
0.1uF
C22
C21
0.1uF
+1V8+1V8
Crystal
C15
15p
Y1
Crystal_GND24
R7
1k
15p
C14
XIN
XOUT
MCU
C12
100n
C9
100n
R10
10k
RUN
R3 27.4
R427.4
USB_DPN
USB_DMN
USB_DP
USB_DM
QSPI_SS
QSPI_SDO
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
SWCLK
SWD
TESTEN
DVDD
IOVDD
USB_VDD
ADC_AVDD
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
U1
RP2040
USB-C & 3.3V Regulator
P1
USB_C_Plug_USB2.0
VBUS
CC
VCONN
D-
D+
SHIELD
R15.1k
R25.1k
PWR_FLAG
+5V
C1
10u
C2
10u
U2
AP2112K-3.3
TROUBLESHOOT and RESET PIN & BUTTON
J1
Conn_01x02_Pin
R5
1k
BOOTSEL (connected to flash)
RESET
SW1
RESET BUTTON (RP2040)
Hassan Iqbal
Sheet: /
File: Sensor Board V2.kicad_sch
Title: Sensor Board RP2040
Size: A4
Date: 2025-11-09
Rev:
KiCad E.D.A. 9.0.5
Id: 1/1